

Experiment A - Electron Spin Resonance

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Abstract

This experiment investigates the Electron Spin Resonance (ESR) of a 2,2-diphenyl-1-picrylhydrazyl (DPPH) sample to determine its Landé g-factor. The experiment is based on the Zeeman effect, where an external magnetic field B_0 splits the degenerate energy levels of unpaired electrons, creating an energy gap $\Delta E = g_S \mu_B B_0$. By varying microwave frequencies from 30 MHz to 70 MHz, resonance was identified using a cathode ray oscilloscope in X-Y mode. A linear regression of the magnetic field versus resonant frequency yielded a slope of 0.0373 mT/MHz. The experimental g-factor was determined to be 1.915 ± 0.024 . While this result illustrates the predicted linear relationship between frequency and magnetic field, it deviates from the literature value of 2.0036 by 4.42%. This discrepancy is attributed to systemic uncertainties, including sample positioning and magnetic field inhomogeneities within the Helmholtz coils.

1 Background & Objectives

Spectroscopy refers to the study of materials by the analysis of the absorption of electromagnetic radiation by those said materials. The Electron Spin Resonance (henceforth referred to as ESR), also known as Electron Paramagnetic Resonance, experiment falls under the broader purview of spectroscopy and studies materials with unpaired electrons by considering the interaction between the magnetic moment μ_S of the electron and an external magnetic field B_0 .

1.1 The Underlying Physics

ESR relies on the spin property of the electron. Spin refers to the intrinsic angular momentum possessed by the electron.

$$S = \pm \frac{1}{2} h \quad (1)$$

Where S represents the spin of the electron. As apparent from (1), an electron either exists in the $+1/2$ or $-1/2$ spin states. The magnetic moment of the electron depends on its spin.

$$\mu_S = \frac{g_S \mu_B}{h} S \quad (2)$$

Where g_S is the Lande g-factor and μ_B is the Bohr magneton. Substituting the value of S into this we get:

$$\mu_S = \pm \frac{g_S \mu_B}{2} \quad (3)$$

In the presence of a magnetic field, a magnetic dipole gains potential energy.

$$E = \mu \cdot B \quad (4)$$

Thus, when we apply an external magnetic field to our material of interest with unpaired electrons, the previously degenerate (possessing same energy) electrons get split into two different energy levels according to their spins. This is called the Zeeman effect. The electrons gain energies according to (3) and (4) in presence of an external field B_0 .

$$E = \pm \frac{g_S \mu_B}{2} B_0 \quad (5)$$

Thus, an energy difference ΔE is created

$$\Delta E = \frac{g_S \mu_B}{2} B_0 - \left(-\frac{g_S \mu_B}{2} B_0 \right) \implies \Delta E = g_S \mu_B B_0 \quad (6)$$

By bombarding our material with electromagnetic radiation, we can cause 'spin-flips' where electrons in the $-1/2$ spin state absorb energy equal to ΔE and transition to the $+1/2$ spin state (higher energy state).

$$\Delta E = h\nu = g_S\mu_B B_0 \quad (7)$$

Where $h\nu$ is energy of the EM wave.

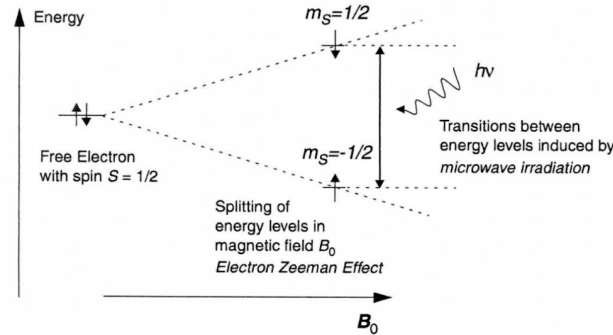


Figure 1: An illustration of the Zeeman effect

The Zeeman effect is depicted in figure 1. Clearly, the stronger the external magnetic field, the greater the energy gap between the two spin states.

**Henceforth, we use f to denote frequency instead of ν .

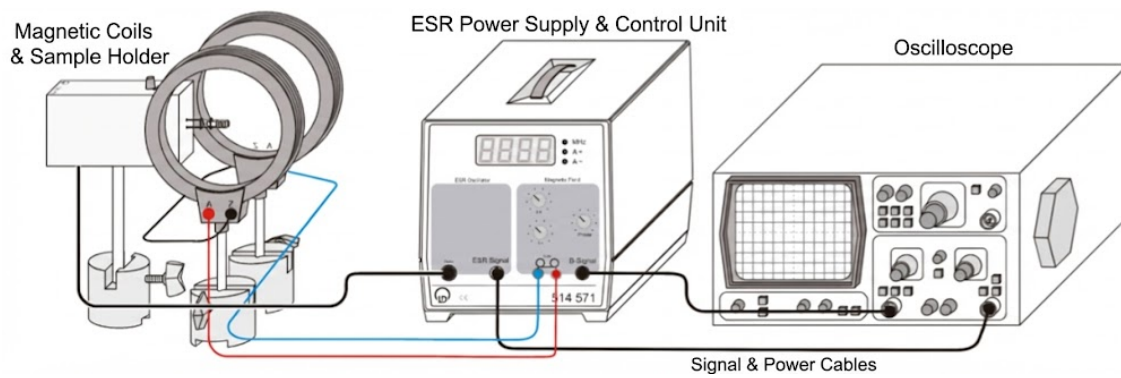
1.2 Objectives

The objective of this ESR set-up is to use the linear relationship between f and B_0 as shown in (7) to determine the g-factor value for our specific sample 2,2-diphenyl-1-picrylhydrazyl or DPPH(see annex).

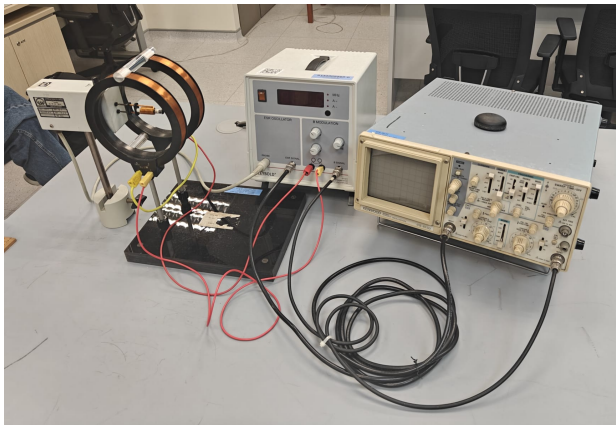
2 Experimental Setup & Methods

2.1 Apparatus

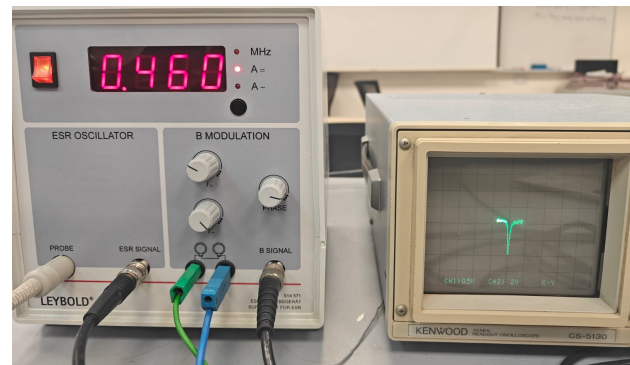
A labeled diagram of the set-up is given below.



(a) ESR Experimental Setup schematic



(b) Experimental set-up



(c) Determination of Resonance

Figure 2: Schematic, apparatus and resonance condition

1. The Helmholtz Coils: These are used to generate the external magnetic fields B_0 and B_{mod} . (more on B_{mod} later)
2. ESR Spectrometer with built in Ammeter: Controls the magnetic field strength and modulation of the magnetic fields.
3. Cathode Ray Oscilloscope: Display used to identify the resonance condition.
4. Sample: DPPH sample
5. High frequency Oscillator: Generates the microwaves that are absorbed by the electrons in the sample at resonance.

2.2 Procedure and Methodology

- The apparatus was turned on and all devices (smartphones, laptops etc.) were taken off the table to prevent noise on the oscilloscope.
- The oscilloscope was set to X-Y mode. In X-Y mode, the impulse on the oscilloscope is time-independent. The Y-axis represents the ESR transmission whereas the X-axis represents the voltage (and thus the magnetic field strength since $V \propto I \propto B$).
- Care was taken to ensure all the wire connections were secured properly. The placement of the sample in the middle of the Helmholtz Coils was checked.

- The frequency of microwave radiation was initially set to 30MHz by adjusting the knob on the high-frequency oscillator. A gap of approximately 10 seconds was taken after setting the frequency to ensure that it reaches a steady-state value.
- B_{mod} was initially set to maximum for a large sweep. B_0 was adjusted till two troughs were seen on the oscilloscope screen. B_0 and B_{mod} were adjusted by using the $A_{=}$ and $A_{\approx}$ knobs respectively on the ESR control unit.
- The phase was adjusted till the two troughs coincided. This setting of the phase was kept constant throughout all rounds.
- B_0 was adjusted again till a symmetric trough was seen. Following this, B_{mod} was decreased to reduce the field sweep and get a more accurate reading.
- The final value of $A_{=}$ was recorded in Ampere by setting the ESR control unit to the appropriate display setting.
- The above procedure was repeated in 5 MHz increments till 70MHz and all the corresponding values of $A_{=}$ were tabulated as raw data.
- B_0 is obtained from the current values using the following relation

$$B = \mu_0 \left(\frac{4}{5} \right)^{\frac{3}{2}} \frac{n}{R} I \quad (8)$$

Here μ_0 is the permeability of free space, n is the number of turns in the Helmholtz coils and R is the radius of the coils.

- A graph of B vs f was plotted and the slope was determined using a line of best fit. The g -factor was then determined using the slope and equation (7).

2.3 Determining Resonance Condition and the role of B_{mod}

By the zeeman effect, the electrons get split into discrete energy levels. Thus, the electrons can only absorb ΔE to transition to a higher energy state. This is very difficult to achieve in practice with a fixed field B_0 . We would observe a resonance impulse only at a specific value of I with negligible margin of error. Thus, B_{mod} , a sweeping field generated by an alternating current, is used in addition to B_0 . The resultant field thus becomes:

$$B = B_0 + B_{mod} \quad (9)$$

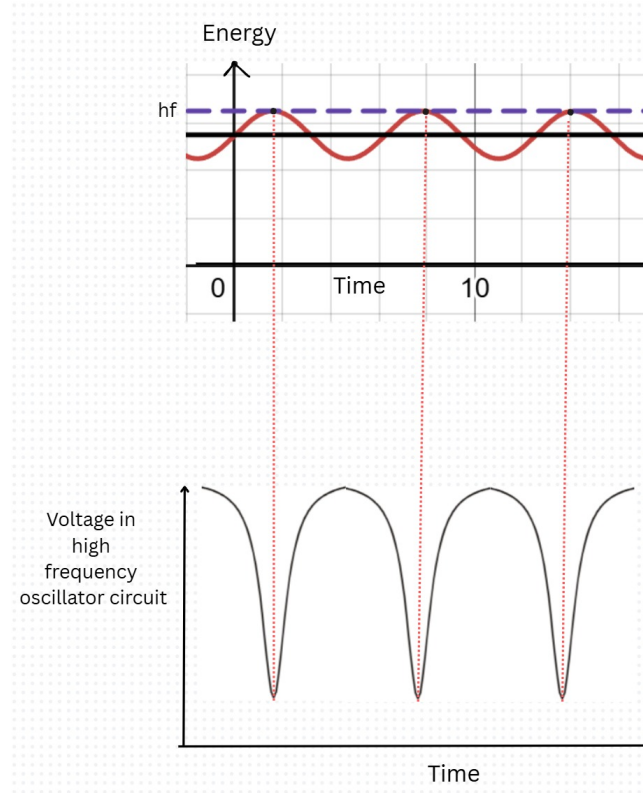


Figure 3: A diagram showing the origin of the troughs on the oscilloscope: The dotted line represents the energy of microwave radiation. The black line represents B_0 whereas the red sinusoid represents the sweep field B_{mod} . When the sinusoid intersects the hf line, resonance occurs and voltage in the high frequency oscillator circuit drops as the electrons absorb energy leading to the troughs. The ultimate goal is to align B_0 with hf . **NOTE: These graphs are purely for the sake of illustration and do not have the appropriate scales, units or axis placement. Additionally, I have neglected the constants from equation (7). Obviously, B is just the magnetic field and not the energy.**

In the time independent X-Y mode, we see the above troughs at resonance.

3 Results & Analysis

3.1 Raw Data

A total of 6 rounds of readings were taken over 3 different days. The first 3 rounds were not considered for analysis (see Annex). The current readings of the last three rounds were averaged to minimise the effect of random errors.

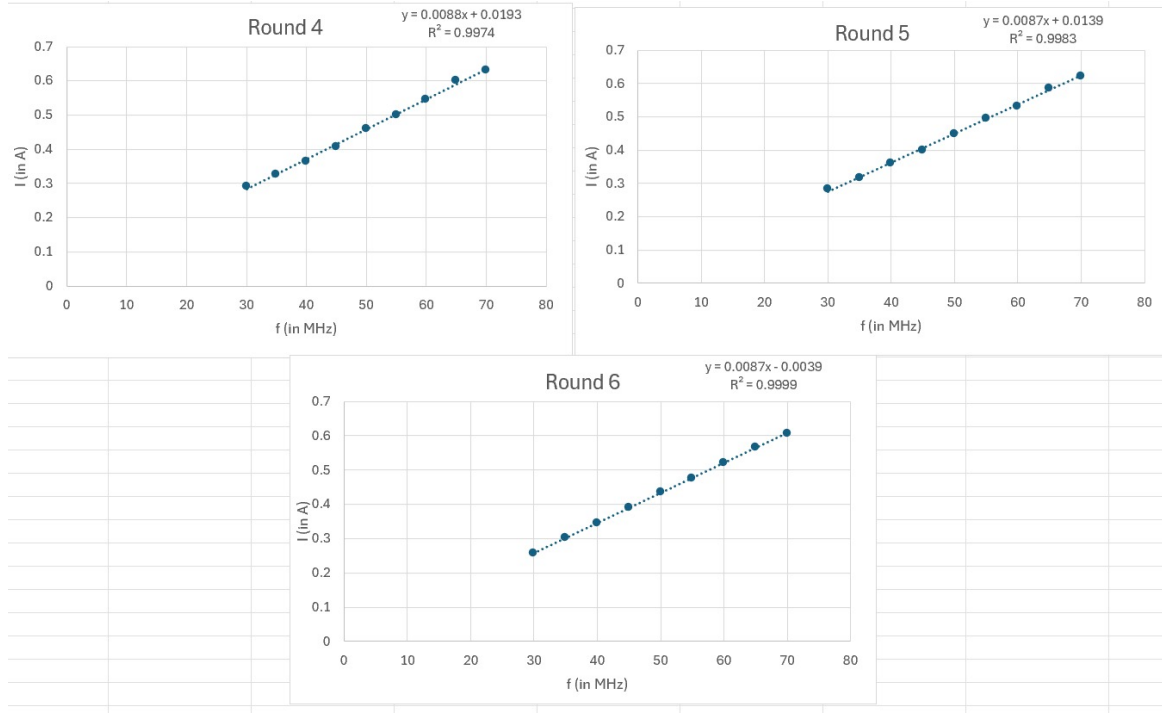


Figure 4: Round-wise I vs f plots of the raw data

We note the value of R squared. As we can see, the raw data already effectively illustrates the linear relationship between B and f since $B \propto I$.

3.2 Analysis of Data

Table 1 lists the frequency, averaged current and corresponding B .

Frequency (in MHz)	I_{avg} (in A)	B_0 (in mT)
30	0.277	1.184
35	0.314	1.341
40	0.357	1.522
45	0.398	1.699
50	0.448	1.911
55	0.491	2.094
60	0.533	2.272
65	0.584	2.492
70	0.620	2.644

Table 1: Values for plotting

We use equation (8) to get B from I taking $n=320$ (given) and $R=0.0675\text{m}$ (measured using a ruler). We graph B vs f and use linear regression to generate a line of best fit. Linear regression is used because it is known from equation (7) that there exists a linear relationship between B and f and this linear relationship has also been illustrated by the plots of the raw data.

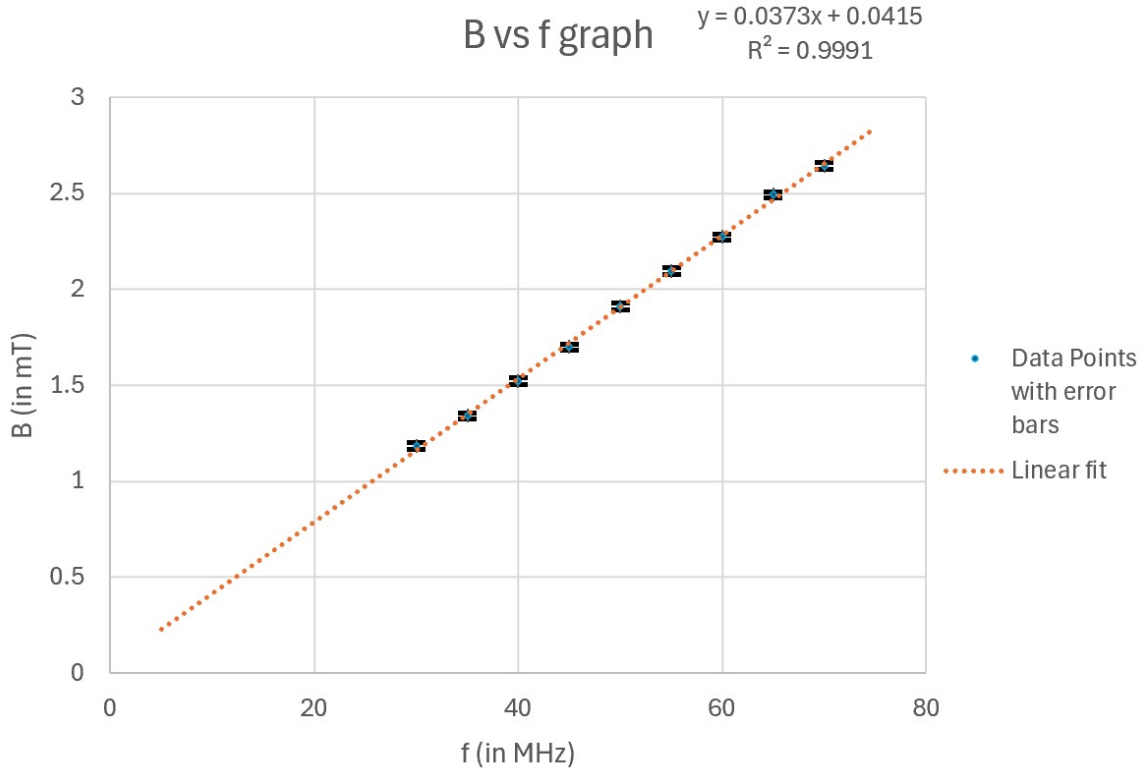


Figure 5: Graph of B vs f with legend

Equation of line of best fit determined through linear regression:

$$B = 0.0373f + 0.0415 \quad (10)$$

Where 0.0373 and 0.0415 are the fit parameters. We also note an R^2 value of 0.9991 which suggests a very strong linear relationship between B and f as predicted by theory. Thus the slope $k = 0.0373 \text{ mT/MHz} = 0.0373 \times 10^{-9} \text{ T/Hz} = \frac{h}{g_S \mu_B}$ (from equation 7).

Using this, we can determine g_S :

$$g_S = \frac{h}{k \times \mu_B} = \frac{6.626 \times 10^{-34}}{0.0373 \times 10^{-9} \times 9.274 \times 10^{-24}}$$

$$\therefore g_S = 1.915 \quad (11)$$

3.3 Error Analysis and Propagation

The error in the final value of the g-factor is determined through error propagation.

The resonance condition is determined manually through sight. Manual error is approximated at 0.5% of I_{mean} . In addition the ammeter reading never reached a steady-state value and fluctuated by $\pm 0.001 \text{ A}$. Finally we need to account for the least count error of 0.001 A. To account for all these errors, error propagation in quadrature was used.

$$I_{mean} = 0.447 \text{ A} \implies \sigma_{I_{manual}} = 0.005 \times 0.447 \text{ A} = 0.002 \text{ A}$$

$$\therefore \sigma_I = \sqrt{0.002^2 + 0.001^2 + 0.001^2}$$

$$\sigma_I = 0.0024\text{A} \quad (12)$$

Now from equation (8), error in B is obtained by considering error propagation of relative errors.

$$\frac{\sigma_B}{B} = \sqrt{\left(\frac{\sigma_R}{R}\right)^2 + \left(\frac{\sigma_I}{I}\right)^2} \quad (13)$$

Here, I have considered the error from known constants negligible. Similar to σ_I , the mean error has been taken as the final value for σ_B and this is the value used for the error bars in figure 5. After calculating the errors individually with σ_R fixed at 0.0005m (L.C/2 of the ruler), and then taking an average:

$$\frac{\sigma_B}{B} = 0.008 = 0.8\% \quad (14)$$

Now the slope is determined as:

$$k = \frac{\Delta B}{\Delta f} \quad (15)$$

Therefore, the slope error σ_k :

$$\frac{\sigma_k}{k} = \sqrt{\left(\frac{\sigma_f}{f}\right)^2 + \left(\frac{\sigma_B}{B}\right)^2} \quad (16)$$

Where $\sigma_f = 0.1 \text{ MHz}$

$$\therefore \frac{\sigma_k}{k} = \sqrt{\left(\frac{0.1}{50}\right)^2 + 0.008^2} = 0.0082 \quad (17)$$

Finally, to calculate uncertainty for the g-factor, we use equation (7) again. From (7),

$$B = \frac{h}{g_S \mu_B} f \implies k = \frac{h}{g_S \mu_B} \quad (18)$$

$$\therefore g_S = \frac{h}{k \mu_B}$$

However, in addition to the slope error, an error of 1% will also be considered to account for error in placement of the sample with respect to the Helmholtz coils and any non-uniformity of the magnetic field over the entire sample due to its finite size.

$$\frac{\sigma_{g_S}}{g_S} = \sqrt{0.01^2 + 0.008^2} = 0.0128 \quad (19)$$

$$\therefore \sigma_{g_S} = 0.0128 \times 1.915 = 0.024 \quad (20)$$

3.4 Final result with uncertainty

Parameter	Determined Value	Uncertainty
g-factor (g_S)	1.915	± 0.024

Table 2: Final result

4 Discussion

The known value of g_S of DPPH is 2.0036. The determined value of $g_S = 1.915 \pm 0.024$ differs from the known value by 4.42%. The known value does not fall within the uncertainty range either. There are several possible reasons for this discrepancy.

1. Placement of the sample - The equations used assume that the sample is placed perfectly at the centre of the Helmholtz coils. Equation (8) gives us the magnetic field at the centre of the coils however, the sample is not a point mass and so it experiences a slightly inhomogeneous field throughout.
2. Systemic uncertainties - Calibration of the various components may have led to errors. In addition, the determination of resonance depended wholly on human perception and thus may lead to subjective errors.
3. DPPH sample itself - The specific sample itself may have a g-factor value that differs from the known value of 2.0036. The g-factor value is known to be affected by the preparation methods of the compound.

Even though the determined value of the g-factor does not match the known value, the experiment has confirmed the linear relationship between B and f through the extremely high value of $R^2 = 0.9991$ which suggests excellent linearity. Thus, the underlying physical laws and principles have been confirmed by the experiment. Future iterations of the experiment can be improved by adding ways to automating the process of resonance detection. The magnetic field strength can also be increased which will lead to a higher ΔE and thus more pronounced troughs.

Presence of a y-intercept in the equation for the line of best fit: The theoretical relationship between the resonant frequency f and the external magnetic field B_0 is given by equation (7), which implies that when the frequency is zero, the magnetic field required for resonance should also be zero. However, the line of best fit for the experimental data yielded a non-zero y-intercept of 0.0415 mT. One primary reason for this is the presence of the Earth's magnetic field and other ambient magnetic fields in the laboratory environment, which can add a constant offset to the field generated by the Helmholtz coils. Additionally, residual magnetism in the cores of the coils or nearby metallic structures may ensure a small magnetic field exists even when the current I is zero. Finally, zero error of 0.004 A was also recorded in the readings of the ammeter which would have also contributed to the presence of the y-intercept.

However, since we are concerned with the slope of the line, the presence of the y-intercept does not change our results.

5 Conclusion

The objective of this experiment was to use the linear relationship between microwave frequency and the external magnetic field to calculate the g-factor of a DPPH sample. Through the observation of the Zeeman effect and the determination of resonance conditions across a range of 30–70 MHz, a clear linear trend was established. The Landé g-factor for DPPH was determined to be 1.915 ± 0.024 . While the result is below the literature value, the linear relationship in the experimental data confirms the fundamental physics of Electron Spin Resonance.

6 References

References

- [1] PC2193 Teaching Team, *Electron Spin Resonance Lab Manual*, 2025.
- [2] *Leybold LD Physics Leaflets, Electron Spin Resonance*
- [3] Zeeman. (n.d.). *Electron Paramagnetic Resonance*. <https://epr.ethz.ch/education/basic-concepts-of-epr/one-elect-in-the-magn-field/zeeman.html>

7 Annex

7.1 DPPH

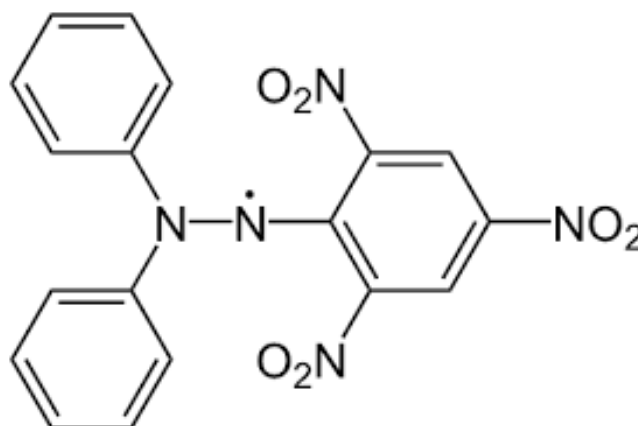


Figure 6: Chemical structure of DPPH

2,2-diphenyl-1-picrylhydrazyl (DPPH) is a dark-colored crystalline powder composed of stable free radical molecules. In most substances, electrons exist in pairs with opposing spins, causing their magnetic moments to cancel out. However, DPPH contains a delocalized unpaired electron centered on the nitrogen atom, making it paramagnetic and an ideal standard for Electron Spin Resonance (ESR) experiments.

7.2 Exclusion of first 3 rounds of raw-data

The first three rounds of data were taken with significant manual error as the B_{mod} field was not modulated correctly leading to inaccurate results. These manual errors were fixed in the subsequent rounds by considering the resonance condition more carefully. Thus, it would be incorrect to give equal weightage to these flawed rounds by including them in the average. Only the three best attempts (4th, 5th and 6th rounds) were included in the report.

8 AI Usage Declaration

I declare that I have used AI in the preparation of this assignment. Prompt: "create a full latex template code complete with hyperlinks in the contents page to all the different sections with structure given in the images. Add a way to check the word count." (the images of DOs and DONTs and report structure were uploaded along with the prompt) Response: Gemini created a full LaTeX code template which I used to make the report.

In general, AI was used for help with LaTeX codes, templates and troubleshooting when LaTeX code did not compile.